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# GAAT: Group Adaptive Adversarial Training to Improve the Trade-Off Between Robustness and Accuracy

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Adversarial training is by far one of the most effective methods to improve the robustness of deep neural networks against adversarial examples. However, the trade-off between robustness and accuracy is still a challenge in adversarial training. Previous methods used adversarial examples with a fixed perturbation budget or specific perturbation budgets for each example, which is inefficient in improving the trade-off and lacks the ability to control the trade-off flexibly. In this paper, we show that the largest element of logit, $z_{\max}$, can roughly represent the

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minimum distance between an example and its neighboring decision boundary. Thus, we propose group adaptive adversarial training (GAAT) that divides the training dataset into several groups based on $z_{\max}$ and develops a binary search algorithm to determine the group perturbation budgets for each group. Using the group perturbation budgets to perform adversarial training can fine-tune the trade-off between robustness and accuracy. Extensive experiments conducted on CIFAR-10 and ImageNet-30 show that our GAAT can achieve a more perfect trade-off than TRADES, MMA, and MART.

*Keywords*: Trade-off; adversarial training; perturbation budget; adaptive.

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## 1. Introduction

Deep neural networks (DNNs) have shown perfect performance in image classification,[22] speech recognition,[18] and natural language processing.[21] However, a growing body of research shows that DNNs are highly vulnerable to adversarial examples.[4,17,32,45] The so-called adversarial examples are those added by carefully designed small perturbations, which are almost imperceptible to humans. Although humans cannot be fooled by adversarial examples, DNNs will make wrong predictions on them.[45] This lack of robustness exhibited by DNNs to adversarial examples has raised serious concerns for security-critical applications. To defend against adversarial examples, many methods have been proposed,[10,19,25,27,36,53] and one of the most effective methods is adversarial training (AT).[17,27]

Standard adversarial training (SAT) uses adversarial examples with a fixed perturbation budget $\epsilon$ to train the network. Although SAT can enhance robustness, it ignores the fact that every example may have different intrinsic robustness, i.e. different distances to the classifier's decision boundary. As such, SAT will change the decision boundary to a great extent, which may seriously affect the accuracy of clean examples at the same time. Previous works show that the robustness and clean accuracy are contradictory.[44,47,53] Balaji *et al.*[3] proposed instance adaptive adversarial training (IAAT) to improve the robustness–accuracy trade-off by setting specific perturbation budgets $\epsilon_i$ for each example. However, for examples close to the decision boundary, IAAT is prone to use adversarial counterparts with smaller perturbations to train a model, which may reduce its robustness. On the contrary, for examples far from the decision boundary, overlarge perturbations are adopted, which is not necessary because the attacker will not choose those examples with such large perturbations in real scenarios. Figure 1 depicts the design of perturbation budgets for SAT and IAAT.

Recent theoretical results[16,28,40] describe this trade-off between robustness and clean accuracy. In practice, different application scenarios need different levels of robustness and accuracy. For example, an autonomous agent might choose to perceive and behave more cautiously, i.e. to prioritize improving its robustness, when it is placed in less confident or adverse environments. So, a flexible adjustment between robustness and accuracy is important. However, SAT and IAAT both lack the flexibility to meet this demand. Therefore, it inspires us to design a better perturbation budget to achieve an optimal trade-off.[47,53] We propose group adaptive adversarial training (GAAT) to achieve flexible tuning of the robustness–accuracy trade-off.

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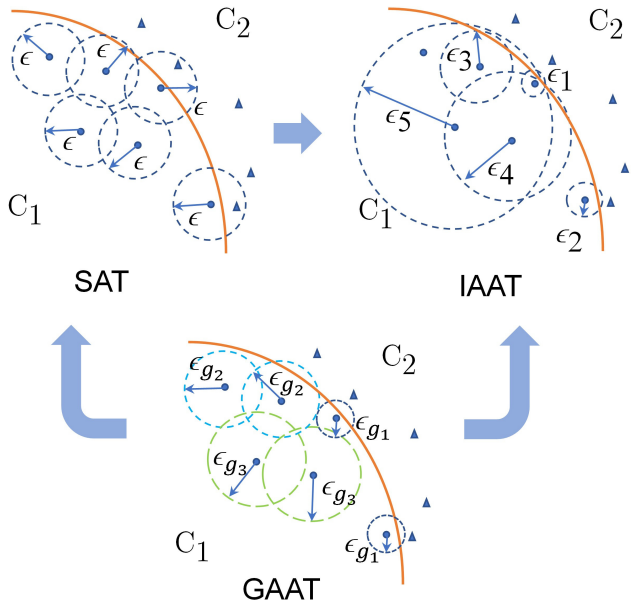


Fig. 1. (Color online) An illustration of different perturbation budgets designed for SAT, IAAT, and GAAT. Here, a red solid line represents decision boundary, a dotted circle represents the range of perturbations, and solid circles and solid triangles, respectively, represent the instances of classes $C_1$ and $C_2$. It shows that GAAT adopted the adaptive perturbation budgets like IAAT, but only has several budgets needed for all examples, which will significantly reduce the computation cost. From another perspective, SAT and IAAT are the special cases of GAAT.

In our work, we use $z_{\max}$, i.e. the largest element of logit, to approximately measure the distance between examples and the decision boundary. By $z_{\max}$, we further divide the data into groups. Then, as shown in Fig. 1, we assign specific budgets $\epsilon_{g_k}$ for each group. That is, every example in the same group shares the same perturbation budget. The number of groups and the group perturbation budgets can be used to fine-tune the trade-off between robustness and accuracy. In particular, SAT and IAAT are two special cases of GAAT. If we view all the training examples as one group, it is equivalent to SAT that merely adopts a fixed perturbation budget. In contrast, if we let each group only contain one example, it becomes IAAT.

Our main contributions are summarized as follows:

- An important fact is revealed that an example with a larger $z_{\max}$ — the largest element of logit — needs a larger perturbation budget to be crafted as an adversarial example. It is proved that $z_{\max}$ is effectively taken as an approximate distance between examples and the decision boundary.
- We propose a group adaptive adversarial training algorithm to flexibly tune the trade-off between robustness and accuracy, which makes adversarial training better suit the needs of scenarios with various security levels.

- A two-stage binary search strategy, including enlarging and shrinking stages, is proposed to acquire more reasonable group perturbation budgets, which is proved to be able to improve the trade-off between robustness and accuracy.
- Extensive experimental results show that GAAT achieves a superior trade-off between robustness and accuracy under both black-box and white-box threat models, when compared to other classic methods.

## 2. Related Works

### 2.1. *Adversarial attacks*

Since Szegedy *et al.*[45] first observed that deep neural networks are highly vulnerable to adversarial examples, this topic has received a lot of attention. Goodfellow *et al.*[17] proposed fast gradient sign method (FGSM) to generate adversarial examples with a single normalized gradient step. It was followed by R+FGSM[46] that adds a randomization step, and the basic iterative method (BIM)[23] that takes multiple smaller gradient steps. These are often grouped under the term projected gradient descent (PGD) which usually refers to the optimization procedure used to search norm-bounded perturbations. Moreover, boundary-based methods like DeepFool[30] and optimization-based methods like CW[7] were also developed, making adversarial defense more challenging. Recently, the ensemble of diverse attack methods — Auto-Attack[12] (AA) by Croce and Hein, consisting of APGD-CE,[12] APGD-DLR,[12] FAB,[11] and Square Attack,[2] became the popular benchmark for testing model robustness. These approaches are also called white-box attacks since the adversary has full knowledge about DNNs.

Except for white-box attacks, there are also many ways to explore the transferability of adversarial examples for black-box attacks. Liu *et al.*[26] was the first to study the transferability of targeted adversarial examples. They observed that a large proportion of target adversarial examples were capable of being transferred with their target labels using the proposed ensemble-based attack method. Dong *et al.*[15] showed that iterative attack methods incorporating the momentum term achieved better transferability. Further, Xie *et al.*[50] boosted the transferability of adversarial examples by creating diverse input patterns with random resize and random padding.

### 2.2. *Adversarial training*

Many of the early adversarial defenses[5,13,37] are broken by stronger attacks. Among the various defense strategies, adversarial training[17,27] currently proves to be the most effective against adversarial attacks,[29,34,39] receiving considerable attention from the research community.

The initial idea of adversarial training is first brought to light in Ref. 45, where neural networks are trained on a mixture of adversarial examples and clean examples. Goodfellow *et al.*[17] went further and proposed FGSM to produce adversarial examples during training. Yet, their trained models remain vulnerable to iterative



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attacks.[46] Mathematically, adversarial training is formulated as a min–max problem. The key to solving this min–max problem is finding strong adversarial examples.[20] As such, Madry *et al.*[27] employed a multi-step gradient-based attack known as PGD attack for solving the inner problem.

Recent advances of adversarial training include adversarial regularization,[17,24,35,48,53] curriculum-based adversarial training,[6,52] ensemble adversarial training,[33,46] adaptively adversarial training,[3,9,14] training with semi-/unsupervised learning,.[1,8,38,51] and efficient adversarial training[41,49] Zhang *et al.*[53] regularized the output from natural images and adversarial examples with the KL divergence function. Wang *et al.*[48] investigated the influence of misclassified examples and proposed Misclassification-Aware adveRsarial Training (MART).

Moreover, Refs. 3 and 14 argued that each example might have different intrinsic robustness. Considering the individual characteristic of adversarial robustness, researchers proposed to do adversarial training at the instance level. Balaji *et al.*[3] first presented the IAAT where $\epsilon$ is selected for each example to ensure that the examples within its $\epsilon$-ball are from the same class. Another work on margin maximization adversarial training (MMA)[14] directly maximizes the margin distance between data points and the model's decision boundary, which is estimated by adversarial perturbations with the least budgets. Adversarial training with adaptive $\epsilon$ is a good exploration, and we think it is promising to improve the trade-off between robustness and accuracy.

## 3. Problem Statement

### 3.1. *Notation*

Let $F(\boldsymbol{x};\boldsymbol{\theta}) : \boldsymbol{x} \in \mathcal{X} \to y \in \mathcal{Y}$ be a DNN mapping the input space $\mathcal{X}$ to the label space $\mathcal{Y} = \{1, \ldots, c\}$. We further denote $\boldsymbol{Z}$ as a logit output of the penultimate layer of the DNN. As known, $\boldsymbol{Z}$ is a vector namely $(z_1, \ldots, z_c)$ and the maximum element is denoted by $z_{\max}$. For a correctly classified input $\boldsymbol{x}$ with the ground-truth label $y$ such that $F(\boldsymbol{x}) = y$, an adversarial example $\boldsymbol{x}'$ is crafted by adding a small perturbation $\boldsymbol{\delta}$ to $\boldsymbol{x}$ without changing the label, but it misleads DNN as $F(\boldsymbol{x}') \neq y$. The adversarial example is presented by $\boldsymbol{x}' = \boldsymbol{x} + \boldsymbol{\delta}$ where the perturbation is constrained by $\|\boldsymbol{\delta}\| < \epsilon$. Here, $\epsilon$ is in general called the perturbation budget. $\|\cdot\|$ is a generic distance norm that can be $l_1$, $l_2$, or $l_\infty$.

To defend adversarial examples, adversarial training is proposed by including adversarial examples into the training procedure, which can be formulated as the following min–max problem[20,27,42]:

$$\min_{\boldsymbol{\theta}} \frac{1}{N} \sum_{(\boldsymbol{x}_i, y_i) \in \mathbb{D}} \max_{\|\boldsymbol{\delta}_i\| \leq \epsilon} L(F(\boldsymbol{x}_i + \boldsymbol{\delta}_i; \boldsymbol{\theta}), y_i), \tag{1}$$

where $\mathbb{D} = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^{N}$ is a training set, $\boldsymbol{x}_i \in \mathcal{X}$ is a clean example, $y_i \in \mathcal{Y}$ is its ground-truth label, $\boldsymbol{\delta}_i$ is the perturbation added to $\boldsymbol{x}_i$, $\epsilon$ is the perturbation budget,

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and $L(\cdot)$ is a loss function. This problem can be regarded as a bilevel optimization problem, where the upper problem is to optimize network parameters $\boldsymbol{\theta}$ to achieve the highest classification accuracy under the strongest adversarial examples, while the lower-level problem is to optimize the perturbation $\boldsymbol{\delta}_i$ to fool the DNN with the highest probability. By optimizing alternately, we can obtain the game equilibrium between the classifier and the attacker. Figure 2 illustrates the entire optimization process. Unfortunately, obtaining the strongest $\boldsymbol{\delta}_i$ is a time-consuming task. In practice, for computation efficiency, some approximate methods such as FGSM,[17] BIM,[24] and PGD[27] are used to generate adversarial examples for adversarial training.

### 3.2. *Robustness–accuracy trade-off*

Previous works show that the robustness and accuracy are contradictory,[44,47,53] i.e. improved robustness will compromise the accuracy of classifiers, and vice versa. SAT can enhance the classifiers' robustness, which imposes the same perturbation budget on all examples. For achieving higher robustness, it tends to use a larger perturbation budget, which may sincerely compromise clean accuracy.

Let $\mathbb{B}(\boldsymbol{x}, \epsilon)$ be a neighborhood of $\boldsymbol{x}$ with the perturbation budget $\epsilon$: $\{\tilde{\boldsymbol{x}} \in \mathcal{X} | \|\tilde{\boldsymbol{x}} - \boldsymbol{x}\| \leq \epsilon\}$. We assume the class of $\boldsymbol{x}$ is $C_1$, and one of its neighbor classes is $C_2$. Thus, the set $\mathbb{S}(C_1, C_2)$ denotes the decision boundary of $C_1$ and $C_2$: $\{\boldsymbol{x} | \exists i, j, i \neq j \wedge z_i(\boldsymbol{x}) = z_j(\boldsymbol{x}) \geq \max_{c \neq i,j} z_c(\boldsymbol{x})\}$. We also use dist $(\boldsymbol{x}, C_1, C_2)$ to represent the distance of $\boldsymbol{x}$ to the boundary $\mathbb{S}(C_1, C_2)$. We take two cases to discuss the influence of fixed perturbation budget $\epsilon$ on the robustness–accuracy trade-off. Table 1 briefly summarizes the influence on examples of SAT using a fixed perturbation budget in the two cases.

For the first case, the example $\boldsymbol{x}$ of $C_1$ is close to the decision boundary such that $\text{dist}(\boldsymbol{x}, C_1, C_2) < \epsilon$, it means the existing examples, e.g. $\tilde{\boldsymbol{x}}$ of $C_2$ within $\mathbb{B}(\boldsymbol{x}, \epsilon)$. An adversarial example $\boldsymbol{x}' = \boldsymbol{x} + \boldsymbol{\delta}$ can be easily crafted such that $F(\boldsymbol{x}') = C_2$. Thus, the clean example $\tilde{\boldsymbol{x}}$ and the adversarial example $\boldsymbol{x}'$ have the same class label $C_2$, and

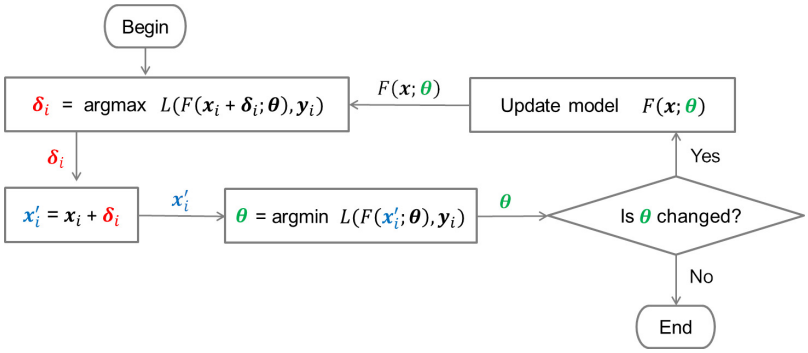


Fig. 2. A flowchart of adversarial training process.

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Table 1. Two cases on the influence of standard adversarial training with a fixed perturbation budget. "Label" represents the ground-truth label of the "Example". "ST" stands for the predicted label by a model with standard training, while "SAT" stands for the predicted label by a model with standard adversarial training. The Y/N under "Influence" indicates whether the standard adversarially trained model exerts an influence on $\boldsymbol{x}$, adversarial example $\boldsymbol{x}'$, and example $\tilde{\boldsymbol{x}}$ within $\mathbb{B}(\boldsymbol{x}, \epsilon)$.

| Case | Example | Label | ST | SAT | Influence |
|---|---|---|---|---|---|
| dist $(\boldsymbol{x}, C_1, C_2) < \epsilon$ | $\boldsymbol{x}$ | $C_1$ | $C_1$ | $C_1$ | N |
| | $\boldsymbol{x}'$ | $C_1$ | $C_2$ | $C_1$ | Y |
| | $\tilde{\boldsymbol{x}}$ | $C_2$ | $C_2$ | $C_1$ | Y |
| $\text{dist}(\boldsymbol{x}, C_1, C_2) \gg \epsilon$ | $\boldsymbol{x}$ | $C_1$ | $C_1$ | $C_1$ | N |
| | $\boldsymbol{x}'$ | $C_1$ | $C_1$ | $C_1$ | N |
| | $\tilde{\boldsymbol{x}}$ | $C_1$ | $C_1$ | $C_1$ | N |

they are also in the same $\epsilon$-ball $\mathbb{B}(\boldsymbol{x}, \epsilon)$. Nevertheless, remind that the goal of adversarial training is to ensure the class label of examples in $\mathbb{B}(\boldsymbol{x}, \epsilon)$ is consistent with the ground-truth label of $\boldsymbol{x}$. For this purpose, adversarial training tends to change the decision boundary to ensure $F(\boldsymbol{x}') = C_1$ and $F(\tilde{\boldsymbol{x}}) = C_1$. Although the robustness is improved (as the adversarial example $\boldsymbol{x}'$ is correctly classified), the accuracy of clean examples is reduced (as the natural example $\tilde{\boldsymbol{x}}$ is misclassified). If we further increase $\epsilon$, there are more clean examples of $C_2$ involved in $\mathbb{B}(\boldsymbol{x}, \epsilon)$ and misclassified.

For the second case, $\boldsymbol{x}$ is far from the decision boundary, i.e. $\text{dist}(\boldsymbol{x}, C_1, C_2) \gg \epsilon$, which means that all the examples in the $\epsilon$-ball $\mathbb{B}(\boldsymbol{x}, \epsilon)$ have a consistent class label. In this case, a perturbed example $\boldsymbol{x}'$ is not really an adversarial example. Unless we increase $\epsilon$ satisfying dist $(\boldsymbol{x}, C_1, C_2) < \epsilon$, we can just craft an adversarial example. As such, Balaji *et al.*[3] proposed an adaptive method, named IAAT, to set a specific perturbation budget for each input example. Unfortunately, it mainly guarantees clean accuracy while compromising the robustness at some level.

### 3.3. *Motivation*

Although IAAT takes into account the different intrinsic robustness levels of each example and provides specific perturbation budgets, it has three shortcomings: (1) too small budgets for the examples nearby the decision boundary will not contribute to improving robustness; (2) too large budgets for the examples far from the decision boundary are not necessary since an attacker will not use such a large budget; and (3) it is time-consuming to determine specific perturbation budgets for each example at each epoch.

In our work, we aim to provide a flexible way to tune the trade-off between robustness and accuracy. Instead of a fixed perturbation budget in SAT or specific budgets in IAAT, we set budgets for each group based on the distance to the decision boundary. Unfortunately, obtaining a precise distance is nontrivial and also time-consuming. We notice that $z_{\max}$ can roughly reflect the minimum distance between an example and its neighboring decision boundary as shown in Fig. 3. Its benefit

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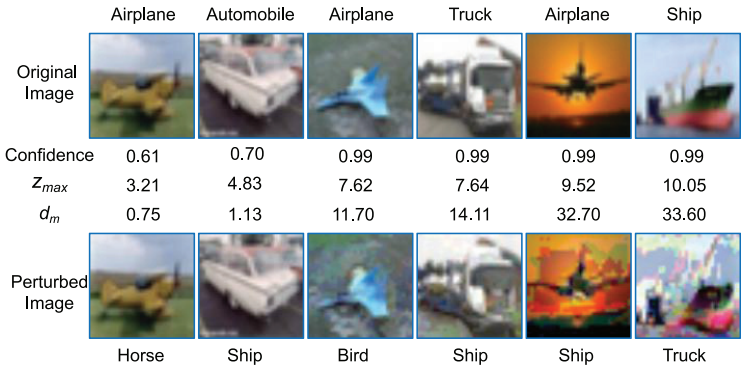
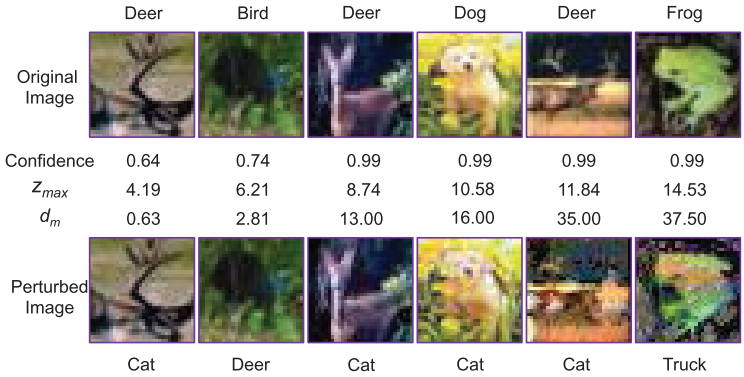
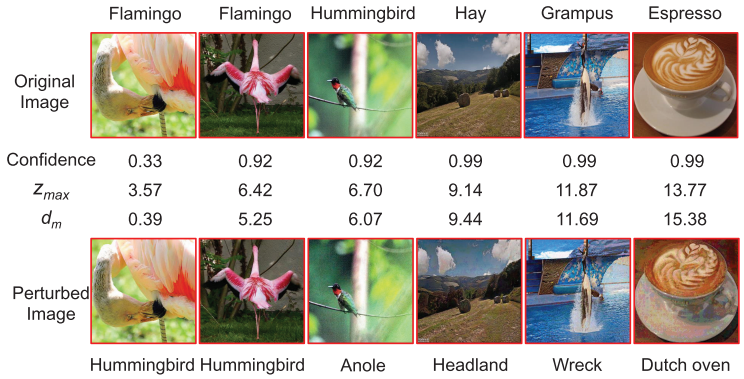



(a) ResNet-18 on CIFAR-10.




(b) WRN-34-10 on CIFAR-10.




(c) ResNet-50 on ImageNet-30.

Fig. 3. Visualizing sampled original images and their perturbed images with different $z_{\max}$. The top row is the original image instances correctly classified by an adversarially trained model. In the middle rows, we present their $z_{\max}$ values in ascending order, and also their corresponding classification confidence values. Here, $d_m$ is the minimum perturbation size required for a successful FGSM attack, which is scaled up by 255. The bottom row is the perturbed images visualized after adding perturbations with a perturbation size $d_m/255$. Intuitively, $z_{\max}$ increases with $d_m$ even though the confidence has reached the same level.

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is that $z_{\max}$ can be directly obtained after forward-propagation without extra computation.

Meanwhile, we notice that the examples with similar $z_{\max}$ also have almost similar $\epsilon_i$, which inspires us to group examples according to their $z_{\max}$. Thus, we assign an adversarial budget $\epsilon_{g_k}$ to a group instead of to an example. That is to say, each example in the same group has the same perturbation budget. IAAT intends to make $\epsilon_i$ as small as possible nearby the decision boundary, while SAT intends to take a large fixed $\epsilon$ for high robustness, thus, in our work, we take a reasonable $\epsilon_{g_k}$ such that $(1/N)\sum_i \epsilon_i < \epsilon_{g_k} < \epsilon$ toward better robustness–accuracy trade-off. In Sec. 4, we will describe GAAT in detail to implement this idea.

## 4. Group Adaptive Adversarial Training

We will first describe the training algorithm at a high level. For obtaining a better robustness–accuracy trade-off, we divide the examples into several groups according to $z_{\max}$ and set specific adversarial budgets for each group in adversarial training. The number of groups, the group limit, and the group perturbation budget are three crucial parameters in our approach. We discuss how to determine the three parameters of GAAT in Secs. 4.2 and 4.3, respectively.

### 4.1. *Method overview*

We train a model with clean examples and adversarial examples at the same time. Formally, GAAT is aimed at solving the following optimization problem:

$$\min_{\boldsymbol{\theta}} \frac{1}{N}\left\{\sum_{(\boldsymbol{x}_i,y_i)\in\mathbb{D}^-} L(F(\boldsymbol{x}_i;\boldsymbol{\theta}),y_i) + \sum_{\mathbb{G}_k\subseteq\mathbb{D}^+}\sum_{(\boldsymbol{x}_i,y_i)\in\mathbb{G}_k}\max_{\|\boldsymbol{\delta}_i\|\le\epsilon_{g_k}} L(F(\boldsymbol{x}_i+\boldsymbol{\delta}_i;\boldsymbol{\theta}),y_i)\right\}, \quad (2)$$

where $\mathbb{D}^+ \subseteq \mathbb{D}$ is a subset of examples correctly classified by the model, $\mathbb{D}^- \subseteq \mathbb{D}$ is a subset of examples misclassified by the model, $\mathbb{G}_k$ $(k = 1, 2, \ldots, K)$ is a data group such that $\cup\mathbb{G}_k = \mathbb{D}^+$ and $\cap\mathbb{G}_k = \emptyset$, $\boldsymbol{\delta}_i$ is the specific perturbation added to $\boldsymbol{x}_i$, and $\epsilon_{g_k}$ is the perturbation budget imposed on $\boldsymbol{x}_i \in \mathbb{G}_k$. All examples in $\mathbb{G}_k$ share the same perturbation budget $\epsilon_{g_k}$. The first term in the optimization problem (2) is standard training with clean examples, while the second term is adversarial training with adversarial examples.

Assume that we run $T$ epochs to complete training. At each epoch, we construct two training subsets $\mathbb{D}^+$ and $\mathbb{D}^-$ where $\mathbb{D}^+ \cup \mathbb{D}^- = \mathbb{D}$ and $\mathbb{D}^+ \cap \mathbb{D}^- = \emptyset$. At epoch $T_i$, we will obtain a new $\mathbb{D}^+$ including the examples correctly classified by the current model, and $\mathbb{D}^-$ including those misclassified. The examples in $\mathbb{D}^+$ are used to generate perturbed examples with $\epsilon_{g_k}$ for adversarial training, while the examples in $\mathbb{D}^-$, which are inherently difficult to classify, are used for standard training. Note that $\mathbb{D}^+$ and $\mathbb{D}^-$ are reconstructed at each epoch since the model's parameters are updated as well. Compared with problem (1), problem (2) is still a bilevel optimization

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problem. Meanwhile, by introducing $\mathbb{D}^+$, $\mathbb{D}^-$, and $\epsilon_{g_k}$, we hope to optimize $\boldsymbol{\theta}$ and $\boldsymbol{\delta}_i$ to obtain a model that classifies well with both the clean and adversarial examples. Therefore, using both more reasonably perturbed examples and clean examples during training ensures good robustness and accuracy.

As described in Sec. 3, we divide $\mathbb{D}^+$ into several data groups $G_1, G_2, \ldots, G_K$ according to $z_{\max}$ of examples. After all examples in $\mathbb{D}^+$ are grouped, the corresponding perturbation budgets $\epsilon_{g_1}, \epsilon_{g_2}, \ldots, \epsilon_{g_K}$ are calculated for each group. We will demonstrate the details in the next two subsections.

### 4.2. *Setting groups*

In our work, we will divide the correctly classified examples into $K$ groups according to $z_{\max}$, and set the perturbation budgets $\epsilon_{g_k}$ $(k = 1, 2, \ldots, K)$ for each group. Here, $K$ is a critical hyperparameter to control the trade-off between robustness and accuracy. If $K = 1$, which means that all clean examples involved in one group share the same budget $\epsilon$, then it degrades into SAT. If $K = |\mathbb{D}^+|$, which means that each group consists of only one example, then it becomes IAAT that tends to guarantee clean accuracy. We choose $1 < K < |\mathbb{D}^+|$ to flexibly tune the trade-off. Under the condition that the robustness–accuracy trade-off is similar, we will select a smaller value as the number of groups, which is convenient for the subsequent process.

Once $K$ is determined, then all we need to do is to set the range of $z_{\max}$ corresponding to each group, i.e. the group limit $E_i (i = 1, \ldots, n)$ such that $G_1 = \{(\boldsymbol{x}_i, y_i) | z_{\max}(\boldsymbol{x}_i) \leq E_1\}, G_2 = \{(\boldsymbol{x}_i, y_i) | E_1 < z_{\max}(\boldsymbol{x}_i) \leq E_2\}$, and $G_3 = \{(\boldsymbol{x}_i, y_i) | z_{\max}(\boldsymbol{x}_i) > E_2\}$. We will use a statistical method to determine $E_i$. First, we train a robust model on the training set. Then we obtain the empirical distribution of $z_{\max}$ of correctly classified examples by the model as shown in Fig. 5. The distribution is more likely to be a skewed distribution than a normal distribution, where most of the examples are clustered on one side of the mean of $z_{\max}$. Suppose the mean of $z_{\max}$ is $\mu$, the variance is $\sigma$, we can set our group limits as $\{\mu \pm a_m \sigma\}$, where $a_m \in \{0, 0.5, 1, 1.5, 2, 2.5, 3\}$, and $[\cdot]$ denotes a rounding operator. In our work, $E_i$ is another parameter to influence the trade-off.

### 4.3. *Perturbation budgets for grouped examples*

Our kernel idea is to assign appropriate perturbation budgets for each group, not for each example. The group perturbation budgets can be determined by prior knowledge, but in this paper, it is determined by the average perturbation budgets of some representative examples in a group. We use a simple binary search as shown in Algorithm 2 for $\epsilon_i$ of each example $\boldsymbol{x}_i$, which consists of two main steps: enlarging and shrinking. An interval $[\epsilon_l, \epsilon_r]$ is first initialized. In the enlarging stage, we update $\epsilon_r \leftarrow \epsilon_r + \varphi$ until $\boldsymbol{x}_i + \mathrm{PGD}(\boldsymbol{x}_i, y, \epsilon_r)$ is misclassified by the model. To avoid an overlarge value assigned to $\epsilon_r$, in the next shrinking stage, the range of $[\epsilon_l, \epsilon_r]$ will be updated by $\epsilon_r \leftarrow (\epsilon_l + \epsilon_r)/2$ until $\epsilon_r - \epsilon_l \leq \gamma$. Finally, $\epsilon_i \leftarrow \epsilon_r$ and the mean

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**Algorithm 1.** Group adaptive adversarial training

**Input:** Epoch number $T$ and minibatch size $M$.
**Output:** $\boldsymbol{\theta}$.

**for** $t = 1$ to $T$ **do**
  **for** $r = 1$ to $\lceil |\mathbb{D}|/M \rceil$ **do**
    $\mathbb{D}^+ \leftarrow \varnothing, \mathbb{D}^- \leftarrow \varnothing,$
    **for** $i = 1$ to $M$ **do**
      **if** $F(\boldsymbol{x}_i) = y_i$ **then**
        Determine $\mathbb{G}_k$ for $\boldsymbol{x}_i$ based on its $z_{\max}$,
        $\boldsymbol{\delta}_i \leftarrow \mathrm{PGD}(\boldsymbol{x}_i, y_i, \epsilon_{g_k}),$
        $\mathbb{D}^+ \leftarrow \mathbb{D}^+ \cup \{(\boldsymbol{x}_i, y_i)\},$
      **else**
        $\mathbb{D}^- \leftarrow \mathbb{D}^- \cup \{(\boldsymbol{x}_i, y_i)\},$
      **end if**
    **end for**
    $L' = \frac{1}{M}(\sum_{\mathbb{D}^+} L(F(\boldsymbol{x}_i + \boldsymbol{\delta}_i; \boldsymbol{\theta}), y_i) + \sum_{\mathbb{D}^-} L(F(\boldsymbol{x}_i; \boldsymbol{\theta}), y_i)),$
    $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \nabla_{\boldsymbol{\theta}} L',$
  **end for**
**end for**

**Algorithm 2.** Search perturbation budget $\epsilon_i$

**Input:** Example $(\boldsymbol{x}_i, y_i)$, a constant $a$, search step size $\varphi$, threshold of budget range $\gamma$, upper bound $u$, and a robust model $F_R$.
**Output:** $\epsilon_i$.

$\epsilon_l \leftarrow 0, \epsilon_r \leftarrow a,$
**while** $(F_R(\boldsymbol{x}_i + \mathrm{PGD}(\boldsymbol{x}_i, y_i, \epsilon_r)) = y_i) \quad \& \quad \epsilon_r < u$ **do**
  $\epsilon_l \leftarrow \epsilon_r, \epsilon_r \leftarrow \epsilon_l + \varphi,$
**end while**
$\epsilon_j \leftarrow (\epsilon_l + \epsilon_r)/2,$
**while** $\epsilon_r - \epsilon_l > \gamma$ **do**
  **if** $F_R(\boldsymbol{x}_i + \mathrm{PGD}(\boldsymbol{x}_i, y_i, \epsilon_j)) = y_i$ **then**
    $\epsilon_l \leftarrow \epsilon_j,$
  **else**
    $\epsilon_r \leftarrow \epsilon_j,$
  **end if**
  $\epsilon_j \leftarrow (\epsilon_l + \epsilon_r)/2,$
**end while**
$\epsilon_i \leftarrow \epsilon_r.$

$(1/|G_k|)\sum_{\boldsymbol{x}_i \in G_k} \epsilon_i(\boldsymbol{x}_i)$ is considered as the group perturbation budget of $G_k$. For examples close to the decision boundary, the group perturbation budget $\epsilon_{g_k}$ satisfies $\epsilon_i \leq \epsilon_{g_k} < \epsilon$, while for examples far from the decision boundary, $\epsilon_{g_k}$ satisfies $\epsilon < \epsilon_{g_k} \leq \epsilon_i$, which facilitates the trade-off between robustness and accuracy.

In general, we tend to choose a large $\varphi$ in the enlarging stage to quickly cross the decision boundary. However, this may lead to an overlarge perturbation budget, which will influence the clean accuracy. Hence, a shrinking stage succeeds in ensuring a smaller perturbation budget. Here we assume $K = 3$. Once all perturbation budgets are obtained for each group, i.e. $\epsilon_1$ for $G_1$, $\epsilon_2$ for $G_2$, and $\epsilon_3$ for $G_3$ such that $\epsilon_1 < \epsilon_2 < \epsilon_3$, we can conduct adversarial training as shown in Algorithm 1. Different from IAAT, the rest of the training examples do not require the calculation of specific budgets, but only be simply judged according to their $z_{\max}$. Given an example $\boldsymbol{x}_i$, we can get its $z_{\max}$ and further know which group it belongs to, i.e. $G_k$, as shown in line 6 of Algorithm 1. Then, the group budget $\epsilon_{g_k}$ can be quickly obtained.

## 5. Experiments

### 5.1. *Experimental setup*

**Datasets.** To evaluate the effectiveness of our models on trade-off between robustness and accuracy, we conduct extensive experiments on CIFAR-10[22] and ImageNet-30 datasets. CIFAR-10 consists of 60 000 color images with the size of $32 \times 32$ in 10 classes. There are 50 000 training images and 10 000 test images. ImageNet-30 is a subset with 30 classes extracted from the ImageNet dataset. Each class has 1 300 training images and 50 validation images. In our experiments, we resize the image to $224 \times 224$.

**Training details.** On CIFAR-10, we use the same neural network architecture as in Refs. 48 and 53, i.e. the residual network ResNet-18 and the wide residual network WRN-34-10. For ResNet-18, following MART,[48] the batch size is 128 and the number of training epochs is 100. All models are trained using the SGD optimizer with Nesterov momentum 0.9,[31] weight decay 0.0035, and an initial learning rate of 0.01, which is divided by 10 at the 75th and 90th epochs. The training attack is $\text{PGD}^{10}$ with random start and step budget $\epsilon/4$, while the test attack is $\text{PGD}^{20}$ with random start, perturbation budget $\epsilon$=8/255, and step size $\alpha = \epsilon/10$. For WRN-34-10, the SGD optimizer is used with weight decay 0.0002 and an initial learning rate of 0.1, which are the only differences from the above setting.

For all experiments on ImageNet-30, the pre-trained ResNet-50 provided by PyTorch is retrained and the SGD optimizer with momentum 0.9 and weight decay 0.0005 is adopted. We further employ cyclic learning rates, which can drastically reduce the number of epochs required for training deep networks.[43] A simple cyclic learning rate schedules the learning rate linearly from zero, to a maximum learning rate, and back down to zero and allows architectures to converge to the benchmark accuracy in tens of epochs instead of hundreds.[49] So we set the number of epochs as 30, batch size as 64, and maximum learning rate as 0.03. Similarly, the training



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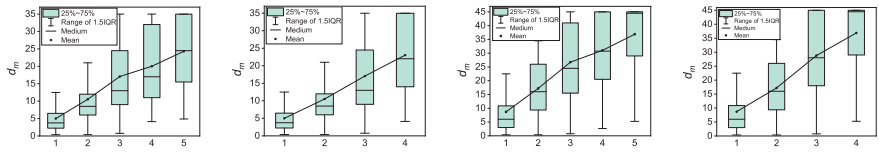
attack is $\text{PGD}^{10}$ with random start and step size $\epsilon/4$, while the test attack is $\text{PGD}^{20}$ with random start, perturbation budget $\epsilon = 4/255$, and step size $\alpha = \epsilon/10$.

### 5.2. *Discussion of grouping*

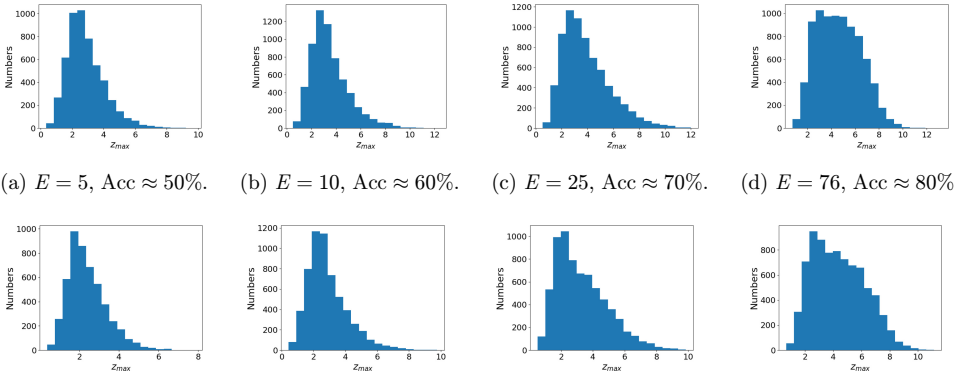
**The reasonableness of grouping.** In Sec. 3, we intuitively demonstrated the relationship between $z_{\max}$ and $d_m$ with sampled images, and assumed $z_{\max}$ is positively related with $d_m$. Based on this assumption, we group examples according to $z_{\max}$ and study the $d_m$ of different grouped examples. As shown in Fig. 4, we observe that as the group index increases, the lower quartile, median, upper quartile, and mean of $d_m$ all increase accordingly. It confirms our assumption that $z_{\max}$ of examples correctly classified by a robust model can approximately measure their minimum distance to the decision boundary. We also notice that grouping with fine-granularity may have a similar distribution of $d_m$, so it is not necessary to construct many groups. Compared with Fig. 4(a), we merge the examples in groups 4 and 5 into one group as shown in Fig. 4(b). Similarly, Figs. 4(c) and 4(d) show the same principle. Therefore, we confirm the reasonableness of $z_{\max}$ to be used for grouping examples. After properly grouping, the $d_m$ values of each group of examples are close enough to share the same perturbation budget. Then we design the group perturbation budgets for each group to achieve a better trade-off between accuracy and robustness.

**Determination of groups.** In Fig. 5, we find that $z_{\max}$ generally presents a right-skewed distribution, and the ranges of their $z_{\max}$ are approximately similar regardless of $\epsilon$. For this reason, the distribution of any $z_{\max}$ in Fig. 5 can be used in designing our grouping scheme. We take one distribution of $z_{\max}$ with a mean $\mu = 4.97$ and variance $\sigma = 1.51$ as the baseline. To avoid few examples in the first or last group, the group upper limit of the first group can be calculated by $[\mu - 1.5\sigma]$, and the group lower limit of the last group is limited to $[\mu + \sigma]$. Other possible group limits between $[3, 6]$ can be $[\mu - 0.5\sigma]$ and $[\mu]$. Here, $[\cdot]$ is a rounding operator. The grouping results are shown in Table 2.




(a) $\epsilon = 4/255$. (b) $\epsilon = 4/255$. (c) $\epsilon = 10/255$. (d) $\epsilon = 10/255$.

Fig. 4. The distribution of $d_m$ of examples in each group. The horizontal axis represents the index of groups. These examples correctly classified by robust models are grouped according to $z_{\max}$. A larger group index indicates its examples with larger $z_{\max}$. Here, two ResNet-18 models ($F_1$ and $F_2$) are adversarially trained with different $\epsilon$ on CIFAR-10. (a) Examples correctly classified by $F_1$ are grouped into five groups. (b) Group 5 of (a) is merged into group 4. (c) Examples correctly classified by $F_2$ are grouped into five groups. (d) Groups 3 and 4 in (c) are merged into one group.




(a) $E = 5$, Acc $\approx 50\%$. (b) $E = 10$, Acc $\approx 60\%$. (c) $E = 25$, Acc $\approx 70\%$. (d) $E = 76$, Acc $\approx 80\%$.

(e) $E = 5$, Acc $\approx 50\%$. (f) $E = 10$, Acc $\approx 60\%$. (g) $E = 30$, Acc $\approx 70\%$. (h) $E = 80$, Acc $\approx 80\%$.

Fig. 5. The distribution of $z_{\max}$ of correctly classified examples by the ResNet-18 models with different accuracy levels on CIFAR-10. We plot the $z_{\max}$ frequency distributions under eight models. (a)–(d) the four models that are trained by SAT with $\epsilon = 8/255$, but resulted from different epochs. Here, different epochs ($E$) represent different accuracy (Acc) levels. (e)–(h) Similarly, the four models that are trained by SAT with $\epsilon = 10/255$, but resulted from different epochs.

Table 2. The grouping results of ResNet-18 on CIFAR-10. $K$ denotes the number of groups, $E_i$ refers to the group limits, and $\epsilon_{g_k}$ is the group perturbation budget.

| | $K$=2 | $K$=3 | $K$=4 | $K$=5 |
|---|---|---|---|---|
| $E_i$ | 3 | 3, 6 | 3, 5, 6 | 3–6 |
| $\epsilon_{g_k}$ | $\epsilon_{g_1} = 5/255$ | $\epsilon_{g_1} = 5/255$ | $\epsilon_{g_1} = 5/255$ | $\epsilon_{g_1} = 5/255$ |
| | $\epsilon_{g_2} = 11/255$ | $\epsilon_{g_2} = 10/255$ | $\epsilon_{g_2} = 9/255$ | $\epsilon_{g_2} = 8/255$ |
| | | $\epsilon_{g_3} = 14/255$ | $\epsilon_{g_3} = 12/255$ | $\epsilon_{g_3} = 10/255$ |
| | | | $\epsilon_{g_4} = 14/255$ | $\epsilon_{g_4} = 12/255$ |
| | | | | $\epsilon_{g_5} = 14/255$ |

**Designing the group perturbation budgets.** Intuitively, the examples with similar $z_{\max}$ have similar potential ability to resist adversarial attack. Thus, the examples in a group can share the same perturbation budget (named group perturbation budget). Figure 6 presents the mean of the examples' perturbation budgets of each group, which is taken as the group perturbation budget. For ResNet-18 on CIFAR-10, the details of group perturbation budgets are summarized in Table 2. In addition, for WRN-34-10 on CIFAR-10 and ResNet-50 on ImageNet-30, the details of group perturbation budgets are shown in Table 3.

### 5.3. *Comprehensive understanding of GAAT*

In this subsection, we analyze the influence of our hyperparameters on the robustness–accuracy trade-off. Besides, we compare models robustness–accuracy trade-offs with different perturbation budget settings.



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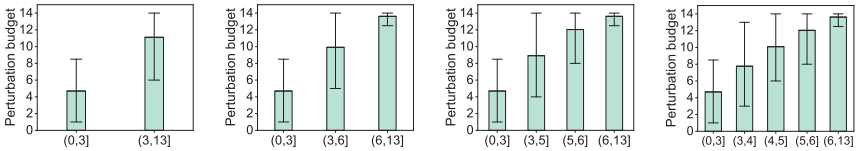


(a) $K = 2$. (b) $K = 3$. (c) $K = 4$. (d) $K = 5$.

Fig. 6. The distribution of the examples' perturbation budget $\epsilon_i$ in each group, where $\epsilon_i$ is obtained through Algorithm 2. Panels (a)–(d) represent four grouping cases, which follow the results in Table 2. In each case, we present the minimum, mean, and maximum values of perturbation budgets for each group of examples. We consider that too small perturbation budgets are not conducive to the models' final robustness, while too large perturbation budgets will reduce the models' clean accuracy. Thus, we set the threshold $\gamma = 1.5$ and the upper bound $u = 14$ in Algorithm 2, to avoid extreme values, meanwhile facilitating the trade-off between robustness and accuracy.

Table 3. The settings of hyperparameters for WRN-34-10 and ResNet-50.

| | | $K = 2$ | $K = 3$ | $K = 4$ | $K = 5$ |
|---|---|---|---|---|---|
| Model | $E_i$ | 3 | 3, 9 | 3, 6, 9 | 3, 5, 6, 9 |
| WRN-34-10 | $\epsilon_{g_k}$ | $\epsilon_{g_1} = 4/255$<br>$\epsilon_{g_2} = 11/255$ | $\epsilon_{g_1} = 4/255$<br>$\epsilon_{g_2} = 10/255$<br>$\epsilon_{g_3} = 14/255$ | $\epsilon_{g_1} = 4/255$<br>$\epsilon_{g_2} = 8/255$<br>$\epsilon_{g_3} = 12/255$<br>$\epsilon_{g_4} = 14/255$ | $\epsilon_{g_1} = 4/255$<br>$\epsilon_{g_2} = 7/255$<br>$\epsilon_{g_3} = 10/255$<br>$\epsilon_{g_4} = 12/255$<br>$\epsilon_{g_5} = 14/255$ |
| | $E_i$ | 3 | 3, 10 | 3, 7, 10 | 3, 6, 7, 10 |
| ResNet-50 | $\epsilon_{g_k}$ | $\epsilon_{g_1} = 2/255$<br>$\epsilon_{g_2} = 6/255$ | $\epsilon_{g_1} = 2/255$<br>$\epsilon_{g_2} = 5/255$<br>$\epsilon_{g_3} = 8/255$ | $\epsilon_{g_1} = 2/255$<br>$\epsilon_{g_2} = 4/255$<br>$\epsilon_{g_3} = 7/255$<br>$\epsilon_{g_4} = 8/255$ | $\epsilon_{g_1} = 2/255$<br>$\epsilon_{g_2} = 4/255$<br>$\epsilon_{g_3} = 5/255$<br>$\epsilon_{g_4} = 7/255$<br>$\epsilon_{g_5} = 8/255$ |

**The adjustment ability of GAAT.** Here, we further analyze the adjustment ability of GAAT regarding the robustness–accuracy trade-off from three aspects: (1) the number of groups $K$, (2) the group limit $E_i$, and (3) the group perturbation budget $\epsilon_{g_k}$.

Table 4. The robust accuracy (RAcc) and clean accuracy (Acc) values of ResNet-18 on CIFAR-10 with different $K$. Here, "Best" is from one epoch that achieves the best robustness–accuracy trade-off throughout the training process, and "Avg" is the average of the last 10 epochs' results. The robust accuracy is based on perturbed test examples by PGD attack with $\epsilon = 8/255$, $\alpha = \epsilon/4$, and step = 20.

| | $K = 2$ | | $K = 3$ | | $K = 4$ | | $K = 5$ | |
|---|---|---|---|---|---|---|---|---|
| | Best | Avg | Best | Avg | Best | Avg | Best | Avg |
| Acc | 84.51% | 83.73% | 84.86% | 84.43% | 85.60% | 85.07% | 85.79% | 85.29% |
| RAcc | 54.74% | 54.13% | 53.89% | 53.38% | 53.46% | 52.73% | 53.41% | 52.71% |

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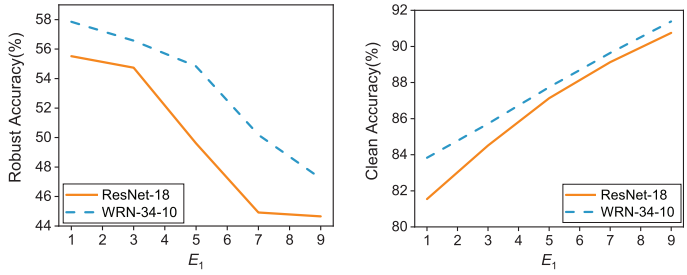
Table 4 shows the robust accuracy and clean accuracy values under different $K$. As $K$ grows, the clean accuracy increases accordingly, while the robust accuracy decreases. It means larger $K$ leads to finer group perturbation budgets, which is inclined to guarantee clean accuracy. Meanwhile, we also find that if $E_i$ and $\epsilon_{g_k}$ of each group are properly set, the model can still achieve high-level robustness and accuracy regardless of $K$. For this reason, $K = 2$ is considered sufficient to achieve a better trade-off.

In Table 5, if the first group perturbation budget $\epsilon_{g_1}$ is fixed, we notice that as the second group perturbation budget $\epsilon_{g_2}$ increases, the robust accuracy in general rises while the clean accuracy falls, and vice versa. It implies that when the examples are properly divided like in Table 2, a larger group perturbation budget helps to improve the robustness but reduces the clean accuracy.

Figure 7 shows the performance of models with different group limits. Obviously, as the group limit $E_1$ increases, the number of examples in the first group increases, and more examples' perturbation budgets drop. This explains why the clean accuracy grows while the robust accuracy decreases. In particular, we find that changing $E_1$ from 3 to 7 sacrifices greater robustness while resulting in only a slight improvement in accuracy. It reflects the necessity and importance of group examples in adversarial training for a good robustness–accuracy trade-off. In summary, three

Table 5. The effect of group perturbation budgets on the final robustness and accuracy of ResNet-18 on CIFAR-10. Here, our assumption that $K$=2 and $E_1$=3 still holds.

| | | $\epsilon_{g_1}$ | | |
|---|---|---|---|---|
| | | 4/255 | 5/255 | 6/255 |
| $\epsilon_{g_2}$ | 10/255 | (85.21%, 53.67%) | (84.83%, 54.59%) | (84.74%, 54.48%) |
| | 11/255 | (84.58%, 54.38%) | (84.51%, 54.74%) | (83.97%, 54.87%) |
| | 12/255 | (83.55%, 54.82%) | (83.48%, 54.98%) | (83.40%, 55.16%) |




(a) RAcc. (b) Acc.

Fig. 7. (Color online) The effect of group limits on the final robustness and accuracy of models on CIFAR-10. The dashed blue lines represent the performance of ResNet-18, while the solid orange lines represent the performance of WRN-34-10. We set $K = 2$, $\epsilon_{g_1} = 5/255$, and $\epsilon_{g_2} = 11/255$ for ResNet-18 and $K = 2$, $\epsilon_{g_1} = 4/255$, and $\epsilon_{g_2} = 11/255$ for WRN-34-10.

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crucial hyperparameters demonstrate the flexibility of GAAT to regulate the robustness–accuracy trade-off, which allows defenders to adaptively cope with more possible scenarios.

**Comparison of perturbation budget settings.** Our GAAT includes two special cases, SAT and IAAT. The most obvious distinction among them is the different perturbation budgets set for training examples. As shown in Table 6, compared to SAT and IAAT, GAAT can guarantee both (1) higher robustness with comparable accuracy and (2) higher accuracy with comparable robustness. Actually, the accuracy and robustness of GAAT-1 are both higher than SAT-10. Similarly, GAAT-2 outperforms SAT-8, GAAT-3 outperforms SAT-6, and GAAT-9 is lightly superior to IAAT. In summary, GAAT achieves a better robustness–accuracy trade-off than IAAT and SAT. Moreover, GAAT can even achieve both higher accuracy and robustness than SAT.

### 5.4. *Robustness–accuracy trade-off*

To further demonstrate the effectiveness of our method, we conduct more extensive experiments on CIFAR-10 and ImageNet-30. We compare GAAT with several classic methods: (1) ST: standard training with clean examples focusing on clean accuracy; (2) SAT[27]: standard adversarial training with perturbed examples to improve the robustness; (3) IAAT[3]: instance adaptive adversarial training to improve the robustness while protecting clean accuracy; (4) TRADES[53]: decomposing the robust error as the sum of the classification error and boundary error to trade robustness off against clean accuracy; (5) MMA[14]: improving the robustness by maximizing the distance from the example to the decision boundary; and

Table 6. The clean and robust accuracies of the models with different perturbation budget settings. Here, RAcc denotes robust test accuracy on perturbed examples under PGD attacks with different iterations and different perturbation budgets. All methods adopt ResNet-18 trained on CIFAR-10. SAT-6, SAT-8, and SAT-10 represent the models trained by SAT with $\epsilon = 6/255$, $\epsilon = 8/255$, and $\epsilon = 10/255$, respectively. GAAT-1, GAAT-3, GAAT-5, and GAAT-9 are the four models obtained by our method with $\epsilon_{g_1} = 5/255$ and $\epsilon_{g_2} = 11/255$, where 1, 3, 5, and 9 are the group limits.

| | | RAcc | | | | | |
|---|---|---|---|---|---|---|---|
| | | Step = 20 | | | Step = 100 | | |
| Method | Acc | $\epsilon = 6/255$ | $\epsilon = 8/255$ | $\epsilon = 10/255$ | $\epsilon = 6/255$ | $\epsilon = 8/255$ | $\epsilon = 10/255$ |
| GAAT-1 | 81.54% | 63.23% | 55.52% | 47.43% | 63.19% | 55.43% | 47.25% |
| SAT-10 | 81.20% | 61.75% | 54.18% | 47.01% | 61.74% | 54.17% | 46.88% |
| GAAT-3 | 84.51% | 63.57% | 54.74% | 45.24% | 63.51% | 53.62% | 45.17% |
| SAT-8 | 83.50% | 61.64% | 52.69% | 43.94% | 61.54% | 52.48% | 43.84% |
| GAAT-5 | 87.13% | 60.56% | 49.60% | 39.51% | 60.41% | 49.39% | 39.25% |
| SAT-6 | 86.81% | 59.94% | 49.50% | 39.15% | 59.90% | 49.31% | 38.81% |
| GAAT-9 | 90.75% | 57.88% | 44.65% | 34.97% | 57.64% | 44.23% | 34.61% |
| IAAT | 90.81% | 55.90% | 44.14% | 34.65% | 55.59% | 43.68% | 34.09% |

(6) MART[48]: investigating the characteristics of misclassified examples to improve the final robustness of the model.

**White-box attacks.** We evaluate models' robustness against four types of attacks, i.e. FGSM, PGD$^{20}$, CW$_\infty$, and Auto-Attack, on two datasets. All attacks have full access to model parameters and are constrained by the same perturbation budget, where $\epsilon = 8/255$ for CIFAR-10 and $\epsilon = 4/255$ for ImageNet-30. Note that the CW$_\infty$ attack denotes using CW-loss within the PGD framework here, which is also with step $= 20$, $\epsilon = 8/255$ ($\epsilon = 4/255$), and $\alpha = \epsilon/10$. Auto-Attack is to reliably evaluate model robustness with an ensemble of diverse strong attack methods. We use the open-source code from Ref. 12 to test our models with the perturbation budget 8/255.

**Black-box attacks.** We verify models' robustness under black-box attacks. We train ResNet-18 on CIFAR-10 and ResNet-50 on ImageNet-30 with different methods. Then we use the well-trained model as the source model or the target model. Adversarial examples are generated from the source model to attack the target model. On both datasets, PGD$^{20}$ (black-box) method is applied to attack various defense models. And the setup follows that specified in the white-box attack.

**Adversarial training settings.** We still follow the basic experimental setup on CIFAR-10 and ImageNet-30 described in Sec. 5.1. Besides, for the training attack of other methods, PGD$^{10}$ is also used with random start and step size $\alpha = \epsilon/4$. The perturbation budget $\epsilon = 8/255$ for CIFAR-10, and $\epsilon = 4/255$ for ImageNet-30. Hyperparameters of the classic training methods are configured as per the original papers: max margin is set to 12/255 (CIFAR-10) for MMA, $\lambda = 6$ for TRADES

Table 7. Comparison of our method with other methods under the white-box attacks on CIFAR-10. We use ResNet-18 and WRN-34-10 for all methods. Acc is the accuracy of clean examples, while RAcc represents the robustness of models.

| | | ResNet-18 | | WRN-34-10 | |
|---|---|---|---|---|---|
| Method | Attack | Acc | RAcc | Acc | RAcc |
| ST | PGD$^{20}$ | 94.01% | 0% | 95.10% | 0% |
| SAT-8 | FGSM | 83.50% | 57.34% | 85.98% | 59.03% |
| IAAT | | 90.81% | 54.94% | 91.89% | 56.23% |
| TRADES | | 81.57% | 56.72% | 84.24% | 60.22% |
| MMA | | 85.10% | 55.72% | 86.13% | 58.73% |
| MART | | 81.39% | 59.68% | 83.60% | 62.45% |
| GAAT | | 84.51% | 58.65% | 85.72% | 61.73% |
| SAT-8 | PGD$^{20}$ | 83.50% | 52.69% | 85.98% | 53.45% |
| IAAT | | 90.81% | 44.14% | 91.89% | 46.33% |
| TRADES | | 81.57% | 52.93% | 84.24% | 54.66% |
| MMA | | 85.10% | 50.47% | 86.13% | 51.57% |
| MART | | 81.39% | 55.33% | 83.60% | 57.89% |
| GAAT | | 84.51% | 54.74% | 85.72% | 56.57% |



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Table 7. (*Continued*)

| | | ResNet-18 | | WRN-34-10 | |
|---|---|---|---|---|---|
| Method | Attack | Acc | RAcc | Acc | RAcc |
| SAT-8 | $CW_\infty$ | 83.50% | 49.23% | 85.98% | 51.05% |
| IAAT | | 90.81% | 41.75% | 91.89% | 43.25% |
| TRADES | | 81.57% | 49.41% | 84.24% | 52.92% |
| MMA | | 85.10% | 47.12% | 86.13% | 49.34% |
| MART | | 81.39% | 49.91% | 83.60% | 52.78% |
| GAAT | | 84.51% | 49.57% | 85.72% | 52.13% |
| SAT-8 | AA | 83.50% | 47.67% | 85.98% | 48.38% |
| IAAT | | 90.81% | 37.94% | 91.89% | 39.53% |
| TRADES | | 81.57% | 48.33% | 84.24% | 50.74% |
| MMA | | 85.10% | 46.17% | 86.13% | 47.02% |
| MART | | 81.39% | 48.03% | 83.60% | 50.78% |
| GAAT | | 84.51% | 47.84% | 85.72% | 50.52% |

Table 8. Comparison of our method with previous methods under the white-box attacks on ImageNet-30. We use ResNet-50 for all methods. Acc is the accuracy of clean examples, while RAcc represents the robustness of models.

| | | ResNet-50 | |
|---|---|---|---|
| Method | Attack | Acc | RAcc |
| ST | $PGD^{20}$ | 96.13% | 0% |
| SAT-4 | FGSM | 87.90% | 71.90% |
| TRADES | | 85.87% | 71.97% |
| MART | | 85.40% | 72.73% |
| GAAT | | 87.83% | 72.60% |
| SAT-4 | $PGD^{20}$ | 87.90% | 68.80% |
| TRADES | | 85.87% | 69.07% |
| MART | | 85.40% | 70.33% |
| GAAT | | 87.83% | 69.53% |
| SAT-4 | $CW_\infty$ | 87.90% | 67.07% |
| TRADES | | 85.87% | 68.47% |
| MART | | 85.40% | 68.07% |
| GAAT | | 87.83% | 68.87% |
| SAT-4 | AA | 87.90% | 62.93% |
| TRADES | | 85.87% | 63.73% |
| MART | | 85.40% | 63.97% |
| GAAT | | 87.83% | 64.60% |

and MART. For GAAT training ResNet-18, we set $K = 2$, $E_1 = 3$, $\epsilon_{g_1} = 5/255$, and $\epsilon_{g_2}$=11/255; when training WRN-34-10, we set $K = 2$, $E_1 = 3$, $\epsilon_{g_1} = 4/255$, and $\epsilon_{g_2} = 11/255$; and when training ResNet-50, we set $K = 2$, $E_1 = 3$, $\epsilon_{g_1} = 2/255$, and $\epsilon_{g_2} = 6/255$.

Tables 7 and 8 show the results of defense models under white-box attacks on CIFAR-10 and ImageNet-30 respectively. For ResNet-18, our method generally

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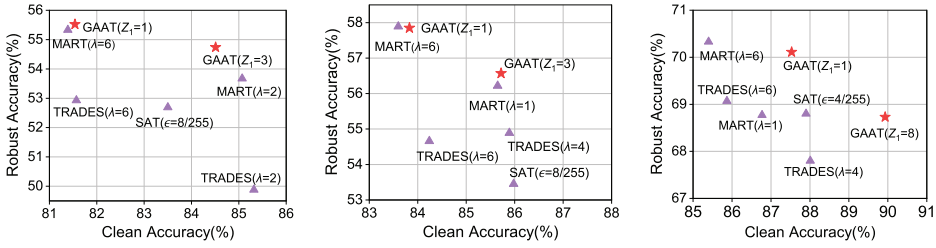
Table 9. Comparison of our method with previous defense models under the black-box attacks on CIFAR-10 and ImageNet-30. Acc is the accuracy of clean examples, while RAcc represents robustness of models.

| Target Model | Acc | RAcc | Source Model | Dataset |
|---|---|---|---|---|
| SAT-8 | 83.50% | 82.74% | ST | CIFAR-10 |
| TRADES | 81.57% | 80.84% | ST | |
| MART | 81.39% | 80.56% | ST | |
| GAAT | 84.51% | 84.27% | ST | |
| TRADES | 81.57% | 62.81% | SAT-8 | CIFAR-10 |
| MART | 81.39% | 63.89% | SAT-8 | |
| GAAT | 84.51% | 63.99% | SAT-8 | |
| TRADES | 81.57% | 62.75% | GAAT | CIFAR-10 |
| MART | 81.39% | 63.88% | GAAT | |
| GAAT | 84.51% | 63.78% | TRADES | |
| GAAT | 84.51% | 66.51% | MART | |
| TRADES | 85.87% | 85.93% | ST | ImageNet-30 |
| MART | 85.40% | 86.33% | ST | |
| GAAT | 87.83% | 87.77% | ST | |
| TRADES | 85.87% | 78.67% | SAT-4 | ImageNet-30 |
| MART | 85.40% | 78.47% | SAT-4 | |
| GAAT | 87.83% | 78.33% | SAT-4 | |
| TRADES | 85.87% | 76.33% | GAAT | ImageNet-30 |
| MART | 85.40% | 77.40% | GAAT | |
| GAAT | 87.83% | 79.67% | TRADES | |
| GAAT | 87.83% | 79.03% | MART | |

achieves both high robustness and accuracy under various attack methods and is the closest to optimal trade-off. Specifically, ST and IAAT have no advantages in trade-off; the robustness of MMA is significantly lower than GAAT under a variety of attacks, while the clean accuracy only improves by 0.59%. Compared with SAT and TRADES, GAAT's robustness and accuracy have been improved at the same time, except for the AA attack. Compared with MART, GAAT's clean accuracy is significantly improved, and its robustness, although slightly lower, is also the closest to MART.

Similar conclusions remain true for WRN-34-10. Particularly, TRADES is slightly more robust than GAAT under the $CW_\infty$ and AA attacks, but the gap is limited within 0.8%, while the clean accuracy of GAAT is improved by 1.48%. Note that although a cyclical learning rate[43] is used in our experiment on ImageNet-30 to reduce training epochs, we find that GAAT still maintains great trade-off ability, and even has better robustness and accuracy than SAT, TRADES, and MART. In addition, we speculate that GAAT plays a better role for high-resolution images.

Table 9 reports the results of black-box attacks. On CIFAR-10, our model is more robust to black-box attacks from different source models, at the same time, its accuracy is significantly higher than other models. Moreover, our model as a source model is capable of generating stronger adversarial examples than TRADES and MART. On ImageNet-30, our model is also more robust than TRADES and MART




(a) (b) (c)

Fig. 8. Comparisons with state-of-the-art methods under PGD[20] (white-box) attack. ★ represents our method, while ▲ stands for other defense methods. We train different models on different datasets: (a) ResNet-18 on CIFAR-10, (b) WRN-34-10 on CIFAR-10, and (c) ResNet-50 on ImageNet-30.

under the black-attack from ST. This demonstrates our method's excellent defense against black-box attacks, even surpassing the most robust MART models in some cases, while still maintaining the highest clean accuracy among these models.

To show the advantages of our methods more intuitively, we plot the comparison results among GAAT, SAT, TRADES, and MART in Fig. 8. We select two representative models for GAAT, TRADES, and MART, respectively, whose key hyperparameter settings (noted in the parentheses) are shown in the figure. It shows that GAAT achieves higher clean accuracy at similar robust accuracy levels, and higher robustness at similar clean accuracy levels, which manifests the good trade-off of GAAT.

## 6. Conclusion

In this paper, we have proposed the GAAT method, to improve the trade-off between robustness and accuracy. We use $z_{\max}$ to group training examples and design the specific group perturbation budgets for each group. The grouping results and the group perturbation budgets play a key role in fine-tuning the trade-off between robustness and accuracy, which allows GAAT to meet various requirements of robustness–accuracy levels under different application scenarios. Experimental results on CIFAR-10 and ImageNet-30 demonstrate the effectiveness of GAAT. Specifically, it can achieve a better trade-off between robustness and accuracy under both white-box and black-box threat models than the other defense methods. Our future work is to extend GAAT to recurrent neural networks and apply it to natural language processing.

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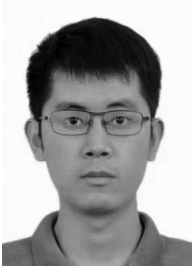

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